# Attention Sink in Transformers: A Survey on Utilization, Interpretation, and Mitigation

Zunhai Su, Hengyuan Zhang, Wei Wu, Yifan Zhang, Yaxiu Liu, He Xiao et al. | LongCat

https://arxiv.org/abs/2604.10098

## Research Questions

- What are the fundamental paradigms for leveraging attention sink (AS) in current Transformer models and how are they applied across different architectures?
- What underlies the emergence and necessity of AS in Transformers and what mechanistic interpretations explain its formation?
- How can future Transformer architectures be designed to operate independently of AS through strategic mitigation approaches?

## Core Idea

Attention Sink (AS) is a phenomenon where Transformers disproportionately focus attention on a small subset of uninformative tokens (like [CLS], [SEP], or initial tokens), creating attention 'sinks' that absorb excess attention mass. This occurs due to softmax normalization constraints forcing attention distribution even when no semantically relevant keys exist for a query. The key insight is that AS serves multiple roles: it can be exploited for efficient inference (sink token preservation), interpreted through mechanistic lenses (outlier circuits, geometric anchoring), or mitigated through architectural modifications (gated attention, modified softmax functions).

## Key Formula

### Attention Sink Detection

$$S_{AS} = \left( i \mid \sum_{j=1}^{L} A_{i,j} > \tau \cdot \mu_A \right), \quad \mu_A = \frac{1}{L} \sum_{k=1}^{L} \bar{A}_k$$

Formal definition of attention sink tokens. S_AS is the set of sink tokens, A_{i,j} is attention weight from token i to token j, τ > 1 is a relaxation threshold (e.g., 1000), μ_A is mean cumulative attention score, and L is sequence length. Identifies tokens whose cumulative attention significantly exceeds the global average.

### Sink Token Preservation

$$\mathrm{Attn}(q_i, K_{J_i}, V_{J_i}) = \mathrm{softmax}\left( \frac{q_i K_{J_i}^T}{\sqrt{d}} \right) V_{J_i}$$

Modified attention computation for sink token preservation. J_i ⊇ Z^sink denotes token indices available to query i, constrained by causality, where Z^sink is the fixed set of sink tokens always available to all queries. Preserves anchor points essential for maintaining model coherence under aggressive compression.

## Key Figure

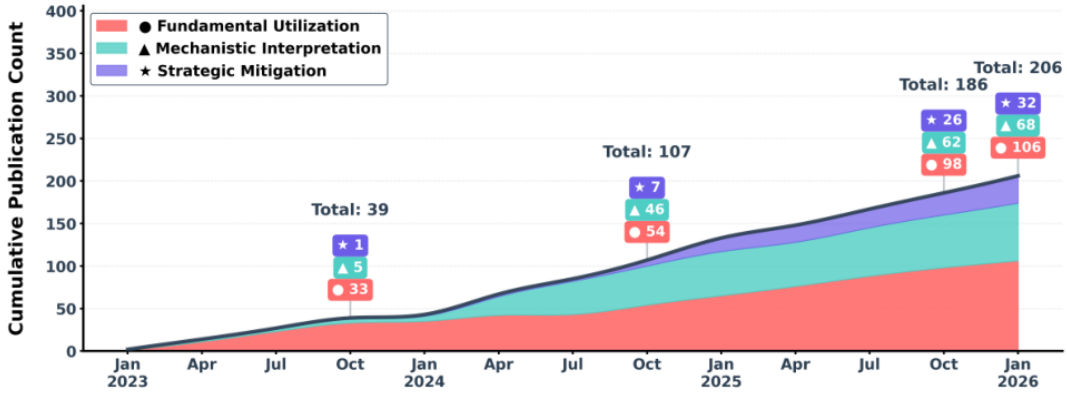


**Figure 3:** Cumulative publication count and temporal trends in AS research from 2023 to 2026. Early research focused on ***Fundamental Utilization*** of AS, followed by studies investigating ***Mechanistic Interpretation***, and most recently, efforts targeting ***Strategic Mitigation*** to address AS and improve model robustness.

### 1.2. Position and Contributions

Building on the preceding analysis, this survey aims to address the lack of a systematic review in AS research. A central focus of our work is to rigorously synthesize AS-related studies across diverse methodologies and Transformer models. By conducting a comprehensive review and taxonomy of over 180 studies, we reveal a dynamic research landscape marked by cumulative progression. As illustrated in Figure 3, the field has steadily broadened its focus: starting from early empirical utilization, the community has progressively incorporated deeper mechanistic understanding and, most recently, systematic mitigation strategies. To capture this multi-dimensional ecosystem, we categorize the literature into three interrelated lines of research:

- **Initial Focus (2023–present) – *Fundamental Utilization*.** Early studies established the empirical utilization of AS [16, 24, 154], emphasizing the exploitation of its inherent characteristics or the management of its immediate effects. This line of research treats AS as a practical phenomenon to be exploited.
- **Deepening Understanding (2024–present) – *Mechanistic Interpretation*.** As empirical applications matured, the community increasingly investigated the underlying causes and architectural factors contributing to AS [27, 43, 98]. This line prioritizes interpretability, aiming for a granular understanding of the internal mechanisms driving the phenomenon and the specific functional roles of AS.
- **Systematic Intervention (2025–present) – *Strategic Mitigation*.** Building on mechanistic insights, the latest research focuses on direct structural mitigation. Studies demonstrate that AS-related extreme tokens can compromise training stability and hinder low-precision deployment [26, 42, 44, 59]. Moreover, misallocated attention to uninformative tokens inherently limits overall model capacity [31, 105, 108, 154]. As a result, developing robust mitigation frameworks has emerged as a critical frontier in current research.

We draw on these three core developmental aspects of AS-related research, presenting our work as the first comprehensive survey of the field. An overview of the survey structure is provided in Figure 1. The detailed section structure is illustrated in Figure 2. Below, we summarize each section:

*Figure 2: Comprehensive organizational structure showing the survey's taxonomy of AS research across different Transformer architectures, utilization methods, mechanistic interpretations, and strategic mitigations with extensive paper citations.*

## Limitations

- Current AS detection methods assume static sink positions, but sinks can dynamically emerge at non-initial positions in some architectures
- Most mechanistic interpretations focus on softmax limitations without fully addressing the complex interaction between attention patterns and model optimization dynamics
- Strategic mitigation approaches often involve architectural trade-offs that may compromise model performance or require extensive retraining
- Limited understanding of AS behavior in emerging architectures like 3D Transformers, hybrid linear attention models, and domain-specific adaptations